

Choices with Risks and Uncertainties

1 Choice Under Uncertainty

1. If an individual is uncertain as to the consequences of his or her actions, we can extend the basic consumer theory to the choice theory with uncertainty.
2. The individual should be able to (a) fully characterize the set of possible consequences; (b) assign a probability to each of these consequences.
3. In short, we are interested in choice over uncertain outcomes. E.g. stock market, weather forecast, R&D, insurance, demand uncertainty, etc.
4. To introduce uncertainty, instead of consumption bundles, the individual will be assumed to have a preference relation over gambles.

1.1 Gambles

1. Possible outcomes:

$$A = \{a_1, a_2, \dots, a_n\}.$$

Each a_i can denote an amount of wealth or a consumption bundle. Here a_i involves no uncertainties.

2. A simple lottery or gamble, denoted g_s , is denoted by

$$(p_1 \circ a_1, p_2 \circ a_2, \dots, p_n \circ a_n),$$

with $p_i \geq 0$ and $\sum_{i=1}^n p_i = 1$. p_i is the probability that the outcome is a_i . Note that here we assume that the probability associated with each outcome is objectively defined and observable.

There is another branch of the theory that deals with subjective probability—situations where the probabilities are not objectively defined.

3. A compound gamble is a gamble whose prizes are gambles (simple or compound). For any compound gambles, we can calculate the probability of each outcome. (I use the term “prize” to refer to what one may receive from a gamble. A prize may be another gamble or an outcome in A .)
4. Example: suppose $A = \{+1, -1\}$.

$$\begin{aligned} g_1 &= (p_1 \circ 1, (1 - p_1) \circ -1) \\ g_2 &= (p_2 \circ 1, (1 - p_2) \circ -1) \\ g_3 &= (q \circ g_1, (1 - q) \circ g_2) \end{aligned}$$

5. We call the simple gamble that lead to the same probability distribution over outcomes as a compound gamble a reduced gamble of the compound gamble. In above, gamble 3 leads to a reduced gamble

$$g_4 = ((qp_1 + (1 - q)p_2) \circ 1, (q(1 - p_1) + (1 - q)(1 - p_2)) \circ -1).$$

6. Axiom: Reduction to Simple Gambles. For any $g, g \sim g_s$, where g_s is the reduced gamble of g . Axiom 6 means that decision makers care only about the probability distribution over outcomes and not the randomizing process.
7. Consequentialism: Ultimately, we derive pleasure from the consumption of the final prize, not the lottery itself. Hence, we should care only about the probability of each prize. But as we shall see, human behavior is not all that simple.
8. Nevertheless, we will always maintain this axiom in the formal analysis. As a result of this axiom, we only need to think about preferences over simple lotteries.

1.2 Axioms on preferences over gambles

1. Completeness. For any two gambles, g and g' , either $g \succsim g'$, or $g' \succsim g$.
2. Transitivity. For any gambles g, g', g'' , if $g \succsim g'$ and $g' \succsim g''$, then $g \succsim g''$.

3. Given Axioms 1 and 2, we can rank all outcomes in A . (a_i can be viewed as a degenerate gamble that yields outcome a_i with probability one.) Without loss of generality, we assume $a_1 \succsim a_2 \dots \succsim a_n$.
4. Axiom 3. Continuity. For any gamble g , there is some probability $\alpha \in [0, 1]$ such that

$$g \sim (\alpha \circ a_1, (1 - \alpha) \circ a_n).$$

5. Axiom 4. Monotonicity. For all probabilities $\alpha, \beta \in [0, 1]$,

$$(\alpha \circ a_1, (1 - \alpha) \circ a_n) \succsim (\beta \circ a_1, (1 - \beta) \circ a_n)$$

if and only if $\alpha \geq \beta$.

6. Axiom 5. Substitution. Consider any $g = (p_1 \circ g^1, \dots, p_k \circ g^k)$ and $h = (p_1 \circ h^1, \dots, p_k \circ h^k)$, where g^i and h^i are gambles. If $g^i \sim h^i$ for all i , then $g \sim h$.
7. Axiom 6: Reduction to Simple Gambles.

2 Expected Utility

1. Definition: The utility function has the expected utility property if, for every gamble g

$$u(g) = \sum_{i=1}^n p_i u(a_i) \tag{1}$$

where $(p_1 \circ a_1, \dots, p_n \circ a_n)$ is the simple gamble induced by g .

2. Utility functions possessing the expected utility property are known as von Neumann-Morgenstern utility functions.
3. Theorem 2.7: Existence of a VNM utility function. Suppose preferences $\succsim$ over gambles satisfy axioms 1-6. Then there exists a utility function u representing $\succsim$ such that u has the expected utility property.
4. We know from classical consumer theory that many utility functions represent the same preferences. Theorem 2.7 says that given our axioms we can always find one version that is linear in probability.

5. Proof: Define $u(g)$ to be the number satisfying

$$g \sim (u(g) \circ a_1, (1 - u(g)) \circ a_n).$$

By Continuity (axiom 3), $u(g)$ exists. By Monotonicity (axiom 4), it is unique. For any g and g' , Monotonicity (axiom 4) implies that $g \succsim g'$ if and only if $u(g) \geq u(g')$. Let $g_s \equiv (p_1 \circ a_1, \dots, p_n \circ a_n)$ be the simple gamble induced by g . By reduction to simple gambles, $g \sim g_s$. Hence, to prove (1), it is sufficient to prove that

$$u(g_s) = \sum_{i=1}^n p_i u(a_i).$$

For each i , $u(a_i)$ satisfies the condition

$$a_i \sim f_{a_i} \equiv (u(a_i) \circ a_1, (1 - u(a_i)) \circ a_n).$$

By Substitution (axiom 5) and reduction to simple gambles,

$$\begin{aligned} g_s &\sim (p_1 \circ f_{a_1}, \dots, p_n \circ f_{a_n}) \\ &\sim \left(\sum_{i=1}^n p_i u(a_i) \circ a_1, \left(1 - \sum_{i=1}^n p_i u(a_i) \right) \circ a_n \right). \end{aligned}$$

Hence, $u(g_s) = \sum_{i=1}^n p_i u(a_i)$.

6. Theorem 2.8. Suppose that VNM utility function u represents $\succsim$. Then the VNM utility function, $v(\cdot)$ represents the same preferences if and only if for some scalar α and some scalar $\beta > 0$,

$$v(g) = \alpha + \beta u(g).$$

Theorem 2.8 says that VNM is invariant to positive affine transformation.

7. Proof: it is straightforward to see that $v(g) \geq v(g')$ if and only if $u(g) \geq u(g')$ when v is a positive affine transformation of u . Suppose both u and v are VNM of $\succsim$. By axioms 3 (Continuity) and 4 (Monotonicity), for any gamble g there exists a unique λ such that

$$g \sim (\lambda \circ a_1, (1 - \lambda) \circ a_n).$$

Hence,

$$u(g) = u((\lambda \circ a_1, (1 - \lambda) \circ a_n)) \quad (2)$$

$$= \lambda u(a_1) + (1 - \lambda) u(a_n) \quad (\text{as VNM is linear}) \quad (3)$$

$$= \lambda (u(a_1) - u(a_n)) + u(a_n) \quad (4)$$

and, similarly,

$$v(g) = \lambda(v(a_1) - v(a_n)) + v(a_n). \quad (5)$$

Rearranging the terms, we have

$$v(g) = \frac{v(a_1) - v(a_n)}{u(a_1) - u(a_n)} u(g) - \left(u(a_n) \frac{v(a_1) - v(a_n)}{u(a_1) - u(a_n)} - v(a_n) \right).$$

Setting

$$\alpha = \frac{u(a_1)v(a_n) - u(a_n)v(a_1)}{u(a_1) - u(a_n)}$$

$$\beta = \frac{v(a_1) - v(a_n)}{u(a_1) - u(a_n)}$$

completes the proof.

8. Theorem 2.8 means that VNM utility function is a cardinal concept—for any u and v that are VNM of $\succsim$, and for any gambles g, g', f and f'

$$\frac{u(g) - u(g')}{u(f) - u(f')} = \frac{v(g) - v(g')}{v(f) - v(f')}.$$

Thus, it is meaningful to say when the ratio is greater than one that a consumer's preference of g over g' is stronger than his preference of f over f' . By contrast, the corresponding ratio has no meaning in classical consumer theory (where we can measure only willingness to trade one good for another).

3 Risk Aversion

1. We now investigate the relationship between a VNM utility function and the agent's attitude toward risks.
2. If $u(\cdot)$ is the agent's VNM utility function, and the simple gamble is $g = (p_1 \circ w_1, \dots, p_n \circ a_n)$, we denote

$$u(g) = \sum_{i=1}^n p_i u(w_i)$$

$$u(E(g)) = u\left(\sum_{i=1}^n p_i w_i\right).$$

3. The individual is said to be

1. risk averse if $u(E(g)) > u(g)$,
2. risk neutral if $u(E(g)) = u(g)$,
3. risk loving if $u(E(g)) < u(g)$.

4. That is, an individual who cares only about the expected value of a gamble and not the distribution of outcomes is risk neutral. In other words, a risk-neutral person is someone who is indifferent between any pair of gambles that have the same expected value.

5. An individual is (strictly) risk averse if for any non-degenerate gamble g (with outcomes given in terms of the person's net wealth level) the individual strictly prefers having the expected value of g with certainty to having g itself.

6. Similarly, we say an individual is risk loving if he prefers a gamble to a certain sum equal to the expected value of the gamble.

7. While both risk-averse and risk-loving behavior exist, the former seems to be more common. Stocks returns are higher than bonds', and insurance companies are making profits selling insurance. On the other hand, risk-loving behavior seems to be mainly confined to lotteries and gambling.

8. Theorem: In the expected utility framework, an individual is risk averse if and only if his VNM utility function u is concave, risk neutral if only if u is linear, and risk loving if and only if u is convex.

9. The theorem follows directly from the Jensen's Inequality: for any gamble $g \equiv (p_1 \circ w_1, \dots, p_n \circ w_n)$,

$$u\left(\sum_{i=1}^n p_i w_i\right) \geq \sum_{i=1}^n p_i u(w_i)$$

if and only if u is concave. Illustrate the Jensen's inequality with gambles with two outcomes.

10. The certainty equivalent (CE) of a gamble g for an individual is defined as the amount of wealth the individual consider to be as good as the gamble g . An individual is risk averse if for all gamble g , the CE of g is strictly less than the expected value of g . The term $P = E(g) - CE$ is known as the risk premium.

11. Example: Suppose a person with a VNM utility function $u(w) = w^{0.5}$ and initial wealth \$10,000 is facing a 50% chance of a \$1900 loss. By definition,

$$\begin{aligned} CE^{0.5} &= 0.5 (10000)^{0.5} + 0.5 (10000 - 1900)^{0.5} \\ &= 0.5 (100) + 0.5 (90) = 95. \end{aligned}$$

3.1 Applications: insurance

1. A strictly risk-averse individual with wealth w is facing a potential loss of D . The probability that the loss will happen is π . The individual must decide how much insurance he wants to purchase. One unit of insurance costs $q < 1$ dollars and pay one dollar in the event of the loss. Let α denote the amount of insurance. The individual's optimization problem is

$$\begin{aligned} \max_{\alpha} \quad & \pi u(w - D + \alpha(1 - q)) + (1 - \pi) u(w - \alpha q) \\ \text{s.t.} \quad & \alpha \geq 0, \alpha \leq w/q \end{aligned}$$

2. Define the Lagrangian function

$$\mathcal{L} = \pi u(w - D + \alpha(1 - q)) + (1 - \pi) u(w - \alpha q) + \lambda(w/q - \alpha).$$

The first-order conditions are

$$\begin{aligned} \pi u'(w - D + \alpha(1 - q))(1 - q) - (1 - \pi) u'(w - \alpha q) q - \lambda &\leq 0, \\ \alpha [\pi u'(w - D + \alpha(1 - q))(1 - q) - (1 - \pi) u'(w - \alpha q) q - \lambda] &= 0 \\ \alpha \geq 0, \alpha \leq w/q, \lambda(\alpha - w/q) = 0, \lambda &\geq 0. \end{aligned}$$

Thus,

$$\begin{aligned} \pi u'(w - D + \alpha(1 - q))(1 - q) &= (1 - \pi) u'(w - \alpha q) q \quad \text{if } \alpha \in (0, w/q), \\ \pi u'(w - D)(1 - q) &\leq (1 - \pi) u'(w) q \quad \text{if } \alpha = 0. \end{aligned}$$

3. Intuitively, purchasing an extra unit of insurance lower utility by

$$u'(w - \alpha q) q$$

in the event that there is no loss and raise the utility by

$$u'(w - D + \alpha(1 - q))(1 - q)$$

in the event the loss occurs. To maximize his expected utility, the individual should purchase an amount such that the ratio of per-unit marginal utility in the two events is equal to the inverse of the ratio of the probabilities of the two events.

4. We call an insurance contract fair if the cost is equal to the expected payout; that is, if $q = \pi$. Suppose this is the case, and assume for the time being that $\alpha \in (0, w)$. Then, the first order conditions requires

$$\pi(1 - \pi)u'(w - D + \alpha(1 - \pi)) = (1 - \pi)\pi u'(w - \alpha\pi).$$

Since u' is decreasing, the above equation holds if and only if

$$w - D + \alpha(1 - \pi) = w - \alpha\pi,$$

or

$$\alpha = D.$$

Since u' is strictly decreasing,

$$u'(w - D + \alpha(1 - \pi)) > (<) u'(w - \alpha\pi)$$

if and only if $\alpha < (>) D$. Thus, only $\alpha = D$ satisfies the first order condition. Furthermore, since u' is strictly decreasing, the optimal α is strictly less than D when $q > \pi$.

5. We have reached two conclusions: A risk-averse individual will (1) purchase full coverage when the insurance is fair and (2) never purchase full insurance when the insurance is unfair. The first result is hardly surprising; the second follows from the fact u is differentiable and, hence, locally linear.

3.2 Demand for a Risky Asset

1. An investor must allocate her wealth w between a risky asset and a safe asset. For simplicity, we normalize the returns to the safe asset to zero. Each dollar worth of the risky asset yields $1 + r_i$ dollar with a probability p_i , $i = 1, \dots, n$. The investor choose $\beta \in [0, w]$, the amount put into the risky asset, to maximize her expected utility. Formally, she must solve the maximization problem

$$\max_{\beta \in [0, w]} \sum_{i=1}^n p_i u(w + \beta r_i).$$

2. Define the Lagrangian function

$$\mathcal{L} = \sum_{i=1}^n p_i u(w + \beta r_i) + \lambda(w - \beta).$$

The first order conditions are

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial \beta} &= \sum_{i=1}^n p_i u'(w + \beta r_i) r_i - \lambda \leq 0, \\ \beta \left[\sum_{i=1}^n p_i u'(w + \beta r_i) r_i - \lambda \right] &= 0 \\ \lambda(w - \beta) &= 0, \quad \lambda \geq 0, \quad \beta \in [0, w].\end{aligned}$$

Alternatively, the conditions can be written as

$$\begin{aligned}\sum_{i=1}^n p_i u'(w + \beta r_i) r_i &= 0 \quad \text{if } \beta \in (0, w), \\ \sum_{i=1}^n p_i u'(w + \beta r_i) r_i &\leq 0 \quad \text{if } \beta = 0.\end{aligned}$$

3. Note that when $\beta = 0$,

$$\sum_{i=1}^n p_i u'(w) r_i = u'(w) \sum_{i=1}^n p_i r_i \leq 0$$

Hence, $\beta = 0$ if and only if the return of the risky asset $\sum_{i=1}^n p_i r_i$ is nonpositive. So we arrive at an important conclusion: An risk-averse investor will not invest in a risky asset with nonpositive rate of return (which means a lower rate of return than the risk-free rate). On the other hand, the investor, no matter how risk averse, will always invest a positive amount on a risky asset with a positive rate of return.

4 Measuring Risk Aversion

1. Given that most economic problems involve uncertainty, it is important that we have a way to measure consumers' risk aversion. For example, automobile manufacturers need to know how risk averse drivers are to decide whether to include a new safety feature and governments need to know how risk averse consumers are to decide whether it is worthwhile to enforce a tighter safety standard.
2. It is quite obvious that in the expected-utility framework, the degree of risk aversion is tied to curvature of the VNM utility function u . [Illustrate this point with a diagram.] However, the second derivative u'' would not be a good measure because it is not invariant to positive-affine transformation. Instead we use the following measure of risk aversion.

3. Definition: The Arrow-Pratt measure of absolute risk aversion is given by

$$\rho_a(w) \equiv \frac{-u''(w)}{u'(w)}.$$

$\rho_a(w)$ measures how a marginal change in w affect $\ln u'(w)$. The Arrow-Pratt measure of relative risk aversion is given by

$$\rho_r(w) \equiv \frac{-u''(w)w}{u'(w)}.$$

It is quite plausible to assume that absolute risk aversion decrease with wealth. We usually assume constant relative risk aversion.

4. The following theorem formalizes the connection between risk aversion and the concavity of u . Theorem: Let $\rho_a(w; u)$ and $\rho_a(w; v)$ denote the Arrow-Pratt measures for VNM utility functions u and v , respectively, and define for all x belonging to the range of v

$$u(w) = h(v(w)).$$

For all w

$$\rho_a(w; u) > \rho_a(w; v)$$

if and only if h is strictly concave.

5. The theorem says that a person is more risk-averse than another (at all wealth level) if and only if his VNM utility is a strictly-concave transformation of the VNM utility function of the other person. Note that h is well-defined as v is monotone.
6. Proof: By construction,

$$u(w) = h(v(w)).$$

Hence

$$\begin{aligned} u'(w) &= h'(v(w)) v'(w), \\ u''(w) &= h''(v(w)) v'(w)^2 + h'(v(w)) v''(w). \end{aligned}$$

Dividing the first equation by the second, we have

$$-\frac{u''(w)}{u'(w)} = -\frac{h''(v(w)) v'(w)}{h'(v(w))} - \frac{v''(w)}{v'(w)}.$$

Since $v'(w) > 0$, $\rho_a(w; u) > \rho_a(w; v)$ if and only if $h(\cdot)$ is strictly concave.

7. Theorem: Suppose individual 1 is more risk averse than individual 2 in the sense that

$$\rho_a(w; u_1) > \rho_a(w; u_2).$$

Then for any gamble g the certain equivalence of g for individual 1 is strictly less than that of individual 2.

8. Proof: From the last theorem, we know that

$$u_1(w) = h(u_2(w))$$

for some strictly concave h .

9. Let CE_1 and CE_2 be the certainty equivalence for individuals 1 and 2, respectively. We have

$$\begin{aligned} u_1(CE_1) &= \sum_{i=1}^n p_i u_1(w_i) \\ &= \sum_{i=1}^n p_i h(u_2(w_i)) \\ &< h\left(\sum_{i=1}^n p_i u_2(w_i)\right) \\ &= h(u_2(CE_2)) \\ &= u_1(CE_2). \end{aligned}$$

10. Returning to the insurance problem. Let α^* denote the optimal insurance for an individual with VNM u . Suppose $q > \pi$ and $\alpha^* \in (0, w)$. Then,

$$\frac{u'(w - D + \alpha^*(1 - q))}{u'(w - \alpha^*q)} = \frac{(1 - \pi)q}{\pi(1 - q)} > 1. \quad (6)$$

Consider another individual with VNM v who is more risk averse than the first individual in Arrow-Pratt sense. It follows from the earlier result that there exists a strictly concave h such that $v(w) = h(u(w))$. Therefore, we have

$$\frac{v'(w - D + \alpha^*(1 - q))}{v'(w - \alpha^*q)} = \frac{h'(u(w - D + \alpha^*(1 - q))) u'(w - D + \alpha^*(1 - q))}{h'(u(w - \alpha^*q)) u'(w - \alpha^*q)}. \quad (7)$$

We know from (6) that

$$w - D + \alpha^*(1 - q) < w - \alpha^*q,$$

and, hence,

$$h'(u(w - D + \alpha^*(1 - q))) > h'(u(w - \alpha^*q))$$

as u is strictly increasing and h is strictly concave. Since

$$\frac{v'(w - D + \alpha(1 - q))}{v'(w - \alpha q)}$$

is decreasing in α , it follows from (7) that for $\alpha \leq \alpha^*$

$$\frac{v'(w - D + \alpha(1 - q))}{v'(w - \alpha q)} > 1.$$

Thus, the optimal insurance for this individual with VNM v must be strictly greater than α^* .

11. We would expect a wealthy person to be less risk averse than a poorer person. Intuitively, the same monetary loss (or gain) means less to the wealthier person. Formally, a VNM utility u exhibits decreasing absolute aversion if $\rho_a(w)$ is decreasing in w . An example of a DARA utility function is $u(w) = \ln w$. More generally, u demonstrates DARA if and only if for any $x_2 < x_1$, $u_2(z) = u(x_2 + z)$ is a concave transformation of $u_1(z) = u(x_1 + z)$.

5 First and Second Order Stochastic Dominance

1. We want to have a formal notion that one lottery is better or more risky than another. There are two natural ways that random outcomes can be compared: according to the level of returns and according to the dispersion of return (risk).
2. A distribution $F(\cdot)$ yielding unambiguously higher returns than $G(\cdot)$ represents the idea of first-order stochastic dominance. When $F(\cdot)$ and $G(\cdot)$ have the same mean, but $F(\cdot)$ is unambiguously less risky than $G(\cdot)$, this is second order stochastic dominance.
3. Definition: For any two lotteries F and G with cumulative distribution function F and G , respectively. We say F first-order stochastically dominates G if for any increasing function u

$$\int u(x) dF \geq \int u(x) dG.$$

4. Proposition. The distribution of monetary payoff $F(\cdot)$ first-order stochastically dominates $G(\cdot)$ if and only if for any x (a real number)

$$F(x) \leq G(x).$$

In words, F f.o.s dominates G if for any x the probability that the payoff of F is less than x is less than the probability that the payoff of G is less than x .

5. Definition. Second order stochastic dominance. For any two lotteries F and G with the same mean, $F(\cdot)$ second order stochastic dominates (or is less risky than) $G(\cdot)$ if for every non-decreasing concave function u we have

$$\int u(x) dF(x) \geq \int u(x) dG(x)$$

6. $F(\cdot)$ s.o.s dominates $G(\cdot)$ if and only if and only if G is a mean preserving spread of F. That is, if y denote the payoff of G and x the payoff of F , there exists some zero-mean lottery H (payoff denoted by z) such that

$$y = x + z.$$

$$E[z] = 0.$$

6 Problems with Expected Utility

1. Allais Paradox (Common Consequence):

g_1 : 1.00 chance of \$1,000,000

g_2 : $\begin{cases} .10 \text{ chance of } \$5,000,000 \\ .89 \text{ chance of } \$1,000,000 \\ .01 \text{ chance of } \$0 \end{cases}$

g_3 : $\begin{cases} .10 \text{ chance of } \$5,000,000 \\ .90 \text{ chance of } \$0 \end{cases}$

g_4 : $\begin{cases} .11 \text{ chance of } \$1,000,000 \\ .89 \text{ chance of } \$0 \end{cases}$

The previous axioms predicts that an individual should prefer g_1 to g_2 if and only if he prefers g_4 to g_3 . In reality, people tend to prefer g_1 over g_2 but g_3 over g_4 .

2. Allais Paradox (Common Ratio):

g_5 : 1.00 chance of \$1,000,000

g_6 : $\begin{cases} .8 \text{ chance of } \$5,000,000 \\ .2 \text{ chance of } \$0 \end{cases}$

g_7 : $\begin{cases} .05 \text{ chance of } \$1,000,000 \\ .95 \text{ chance of } \$0 \end{cases}$

g_8 : $\begin{cases} .04 \text{ chance of } \$5,000,000 \\ .96 \text{ chance of } \$0 \end{cases}$

3. Ellsberg Paradox:

A bag contains 300 balls; 100 are red and the other balls are either blue or green.

- (a) Suppose you have to choose a color. If you draw a ball of that color, you will win \$1000. What color will you choose?
- (b) Suppose you have to choose a color. If you draw a ball not of that color, you will win \$1000. What color will you choose?

4. The St. Petersburg Paradox.

Suppose someone offers to toss a fair coin repeatedly until it comes up heads, and to pay you \$2 if this happens on the first toss, \$4 if it takes two tosses to land a head, \$8 if it takes three tosses (and, generally, $\$2^n$ if takes n tosses). What is the largest sure gain you would be willing to forgo in order to undertake a single play of this game? (Machina 1987)

- (a) The probability of that the first head happens at the n^{th} toss (the first $n - 1^{th}$ tosses are tail and the last one is head) is 0.5^n , and the prize when it happens is 2^n . Hence, the expected value of this gamble is

$$1 + 1 + 1 + \dots = \infty.$$

But clearly, the bet will not be worth more than a few thousands for any normal persons. After all, the probability of getting a prize greater than \$128 is less than 1%.

- (b) If $u(w) = w^{0.5}$, then the expected utility of the bet is only 2.414.

5. Homework. (J&R), 1.27; 1.29; 1.50; 1.51; 1.54; 2.20; 2.25; 2.30; 2.37. Due: Nov. 6, hard copy please.

7 Behavioral Economics

1. Neoclassical economics assume fully rational behavior. Behavioral economics study of economic behavior that departs from full rationality. This branch of research seeks to integrate the insights and methods of psychology into economics. One of the pioneers in this area, Daniel Kahneman, who is in fact psychologist by training, won Nobel prize in economics in 2002.
2. The general theme that connects the findings in behavior economics is that the ability of humans to make payoff-maximizing decisions may be limited. The limits fall into three area: (1) limited cognitive/calculating ability; (2) limited willpower; (3) limited self-interest.
3. The first limit relates to complex decisions or decisions that require some calculations. Decision involving uncertainty, for example, require the person to be able to work with probabilities and expected values. Decisions about investments may require the person to understand formulas for present discounted values. When a person learns new information, whether and how much a person should alter his or her decisions actually involves some complex math if it is to be done in any sense optimally. A real person may make mistakes in performing complex calculations or may avoid the calculations entirely and instead rely on an educated guess.
4. The second limitation is important for dynamic decisions, that is, decisions involving some sort of timing element where actions taken up front may have longer term implications. For example, at the beginning of the week, a student may make plans for how much he will study for a test at the end of the week. When time comes, the lure of video games may be too strong, and he may abandon the plans to study. After, the student may even regret having not studied. Such self-control problems may arise in many contexts including diet, exercise, smoking, saving, and so on.
5. Humans may care not just about the income or consumption levels that they and others end up with. They may get direct utility from broader social goals such as fairness and justice. Whether you want to be kind or nasty to someone else may not be predetermined but might depend on whether they were kind or nasty to you previously.
6. One explanation offered by the behavioral economics for Allais Paradox is that people find it difficult to think through the expected-

value formula. In particular they have trouble when small probabilities are involved, tending to overweight them. This might explain why people tend not to like gamble g_2 if they mistakenly put too much weight on the very slim (1%) chance of getting nothing.

7. In a set of experiments run by Kahneman and Tversky (1979), different group of subjects were presented with one of the following two scenarios:
 - (a) Scenario 1: In addition to \$1000 up front, the subject must choose between two gambles. Gamble A offers an even chance of winning \$1000 or nothing. Gamble B provides \$500 with certainty.
 - (b) Scenario 2: In addition to \$2000 up front, the subject must choose between two gambles. Gamble C offers an even chance of losing \$1000 or nothing. Gamble D results in the loss of \$500 with certainty.
8. The authors found 16% of subjects chose A in the first scenario, and 68% chose C in the second scenario. Although the two scenarios are framed in different ways, the allocations are identical across them. A and C both involve an even chance of gaining \$1000 and \$2000, and B and D both involve a certain total payment of \$1500. Simply changing the way the choices are framed, which should be irrelevant from an economic standpoint, leads people to change their decisions.
9. One explanation is, again, that subjects make mistakes in the difficult calculations involved in decisions under uncertainty. All four gambles A through D provide the same expected return (\$1,500). A risk averse subject should then choose the one with no risk (B or D). The people who chose C over D might have been confused by how the choice was framed and perhaps may not have computed final allocations correctly or at all.

7.1 Prospect Theory

1. Kahneman and Tversky took a different view. Rather than mistakes, the choices in the experiments reflected subjects' legitimate preferences, but preferences that do not fit the standard model. They proposed a new model, called prospect theory, the key ingredient of which is that people are very sensitive to small losses from their current wealth.

2. Aversion to small losses can be modeled by putting a kink in the utility function at the current wealth level (called the reference point R), as in (b) of the Figure 1 (draw the diagram). The function's has a steeper slope to the left of R than to the right, so that small gains cannot compensate for small losses. The utility function flattens out moving further to the left of R, implying that people become less sensitive to large losses. When wealth changes, the utility function shifts so that a new kink forms at the new level of wealth. The new wealth level establishes a new reference point. A person's utility is no longer just a function of final wealth, as with standard utility functions, but a function of the path (of gains and losses) by which he or she arrived at that final wealth.
3. Figure 2 shows how such a utility function can capture the results of the authors' experiment. In the first scenario, drawn in (a), the initial payment of \$1,000 shifts the reference point over to R'. The additional payments (\$1,000 with probability 1/2 with A and \$500 for certain with B) are perceived as gains. The person has standard preferences over gains. Given that both outcomes the subject prefers the sure outcome (B) over the risky one (A).
4. In the second scenario, shown in (b), the larger initial payment of \$2,000 shifts the reference point over even further to the right, so that the additional transfers are now perceived as losses. The certain prospect of losing \$500 is so painful to subjects that they would trade this for a smaller chance of losing a larger amount. This shows up on the graph as point C lying above point D.
5. There is a big debate in the economics literature about how best to approach choice anomalies such as revealed in the experiments of Allais or Kahneman and Tversky. Are these better modeled as mistakes, or are they legitimate, rational preferences that simply require us to rethink what it means to be rational? One empirical test is to see how subjects behave when they are allowed more time to become familiar with the choice setting, try out different choices perhaps, and learn with experience. If the subjects continue to exhibit the anomalous choices, then they are probably legitimate preferences. If they change to the rational choice, then the anomaly might have been a mistake.
6. Kahneman and Tversky's experiment revealed that subjects' decisions can be affected simply by restating the same choice in two different ways. When a choice was expressed as a gain to a small initial endowment, people tended to make different decisions than

when the same choice was expressed as a loss from a larger initial endowment. The phenomenon that small changes in the wording of choices can affect decisions, labeled a framing effect, has been found by psychologists and behavioral economists to apply very generally to many different areas of human decision making. Beef labeled 80% lean is preferred to that labeled 20% fat. Basketball players who make 52% of their shots are judged to be better than those missing 48%. Patients may opt for surgery with a 95% survival while avoiding a procedure with 5% chance of death.

7. Overall, it appears that framing a choice around a positive attribute tends to bias people toward preferring that choice. The existence of framing effects poses a problem in economics because the theory tends to focus on real outcomes and does not have much to say about wording. Here is certainly an area where psychology can contribute to economics. Economists need to better understand such questions as when framing effects are strongest, whether people can be trained through experience to see through framing and how framing effects can best be integrated into standard economic models.